
FlashMind

Local AI Flashcard Generator with Spaced Repetition

DAY 08 | JULY 5, 2026

Paste notes. AI builds the deck. Science does the rest.

Category	Details
Startup Name	FlashMind
Type	Offline AI Flashcard Generator with SM-2 Spaced Repetition (Mobile-Only)
Stack	Vanilla HTML/JS + CSS 3D Transforms + Ollama
Models	llama3.2, mistral, gemma2, qwen2.5
Algorithm	SM-2 Spaced Repetition (simplified)
Privacy	100% Local — Notes Never Leave Device
Author	Mayank Swaraj / Mayank Creations
Series	Daily Local LLM Startup — Day 08
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WHITEPAPER

FlashMind: AI Card Generation + Local Spaced Repetition

Abstract

FlashMind is a mobile-first, offline flashcard tool that uses a local LLM to generate Q&A; cards from any pasted text or topic, then schedules review sessions using a simplified SM-2 spaced repetition algorithm. All card data, scheduling metadata, and review history are stored in browser localStorage — no server, no account, no sync.

1. Problem Statement

Anki, the gold standard for spaced repetition, requires manual card creation — a time barrier that prevents most learners from ever building a useful deck. Cloud alternatives (Quizlet AI, Brainscape) auto-generate cards but transmit note content to external servers. No tool combines local AI generation with offline spaced repetition scheduling in a single zero-install file.

2. Solution

- Paste-to-deck in seconds — LLM generates 5-20 Q&A; cards from any text or topic string
- Difficulty levels — Beginner/Intermediate/Advanced modifies prompt instruction for question depth
- SM-2 scheduling — ease factor, interval, and due-date computed per card after each rating
- 4-button rating — Again/Hard/Good/Easy maps to SM-2 quality scores 1-4
- CSS 3D flip animation — smooth card flip on tap, no library required
- Mastery tracking — cards with reps ≥ 3 and ease ≥ 2.3 counted as mastered per deck
- Due-first queue — study sessions prioritise overdue cards, fall back to first 10 for new decks

3. Technical Architecture

Layer	Technology	Role
UI	Vanilla HTML5/CSS3 (mobile-only)	3-tab: Generate, Study, Decks. Max-width 480px
Card Flip	CSS 3D transform + backface-visibility	rotateY(180deg) transition; no JS animation library
Logic	Vanilla JS (ES6+)	Deck CRUD, SM-2 scheduling, study queue management
Storage	localStorage	Full deck objects including per-card SM-2 metadata
AI Runtime	Ollama local process	JSON-array card generation with count + difficulty constraints
Scheduling	SM-2 Algorithm (simplified)	ease, interval, reps per card; due timestamp in ms

4. Market Opportunity

The global e-learning market exceeded USD 250B in 2024. Spaced repetition tools serve an estimated 30M+ active Anki users alone. The card creation bottleneck is the primary churn driver — users who cannot quickly build relevant decks abandon the tool. FlashMind eliminates this bottleneck entirely. Target segments include students, language learners, medical professionals, and corporate training users who handle sensitive study material requiring local-only processing.

5. Monetization

Tier	Price	Offering
Free	USD 0	Unlimited decks, 10 cards/generation, basic stats
Pro	USD 7/mo	20 cards/generation, deck export, streak tracking, multiple profiles
Lifetime	USD 44	Everything in Pro forever
Education	USD 89/classroom/yr	Shared deck library, teacher dashboard, progress reports

6. Competitive Advantage

Feature	FlashMind	Anki	Quizlet AI	Brainscape
AI Card Generation	YES	NO	YES (cloud)	NO
100% Offline	YES	YES	NO	NO
No Account Required	YES	YES	NO	NO
Spaced Repetition	YES	YES	Limited	YES
Single HTML File	YES	NO	NO	NO

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SOURCE CODE PAPER

Annotated technical breakdown of FlashMind.html

SOURCE CODE PAPER

FlashMind.html — Annotated Technical Breakdown

Overview

FlashMind combines three independent systems in one HTML file: AI card generation via Ollama, CSS 3D flip animation for the study UI, and a simplified SM-2 spaced repetition scheduler. No external libraries are used.

Section 1: CSS 3D Card Flip

The flip animation is pure CSS — no JS animation frames:

```
.flashcard { transform-style: preserve-3d; transition: transform .5s; }
.flashcard.flipped { transform: rotateY(180deg); }
.card-face { backface-visibility: hidden; position: absolute; inset: 0; }
.card-back { transform: rotateY(180deg); }
```

Both faces are absolutely positioned within the same container. The front face is visible by default; the back face is pre-rotated 180deg. When .flipped is added, the container rotates and the backface-visibility rule hides whichever face is pointing away.

Section 2: SM-2 Spaced Repetition Algorithm

Simplified SM-2 implementation per card rating:

```
// quality: 1=again, 2=hard, 3=good, 4=easy
if(quality >= 3) {
  if(reps==1) interval=1;
  else if(reps==2) interval=6;
  else interval=Math.round(interval*ease);
  ease=Math.max(1.3, ease+(0.1-(5-q)*(0.08+(5-q)*0.02)));
} else { reps=0; interval=1; }
due = Date.now() + interval*86400000;
```

The ease factor starts at 2.5 and adjusts per rating. Cards rated Again or Hard reset reps to 0 and interval to 1 day. The due field stores a Unix timestamp in milliseconds, enabling precise scheduling across browser sessions.

Section 3: Study Queue Logic

Due cards are prioritised; new decks fall back to the first 10 cards:

```
var now = Date.now();
studyQueue = deck.cards.filter(c => c.due <= now);
if(!studyQueue.length) studyQueue = deck.cards.slice(0,10);
```

Section 4: Function Reference

Function	Role
generateCards()	JSON-array-constrained Ollama call; injects count+difficulty into prompt; builds deck object
startStudy(deckId)	Builds due-first study queue, sets studyDeckId, calls showNextCard()
showNextCard()	Renders card HTML with front/back faces; disables rating buttons until flip
flipCard()	Toggles .flipped class on .flashcard; enables rating buttons
rate(quality)	SM-2 update on current card; persists to localStorage; advances studyIdx
updateProgress()	Computes done/total ratio, updates progress bar width and label
renderDecks()	Renders deck cards with live due/mastered counts computed from card metadata

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HISTORY PAPER

Origin story, design decisions, and build log

HISTORY PAPER

FlashMind — Origin, Design Decisions & Build Log

Origin Story

FlashMind is Day 8 of the Mayank Creations Daily Local LLM Startup series, built July 5, 2026. The series has now covered all major productivity verticals: personal (Day 1), work (Day 2), AI tools (Day 3), finance (Day 4), dev tools (Day 5), learning (Day 6), communication (Day 7). Day 8 returns to learning — but adds the missing ingredient that makes knowledge actually stick: spaced repetition scheduling.

Design Philosophy

SM-2 in 20 Lines

The full SM-2 algorithm can be reduced to ~20 lines of vanilla JS without losing its core mechanics. Ease factor, interval growth, and due-date stamping are sufficient for meaningful spaced repetition without implementing the full Anki SuperMemo variant.

CSS Flip, No Library

The 3D card flip is achieved with 4 CSS rules — transform-style, backface-visibility, rotateY, and transition. This is Day 8's signature technical contribution: a smooth, physics-feeling interaction with zero JS animation code.

Sky Blue Palette

Sky blue (#38bdf8) evokes clarity, knowledge, and focus — appropriate for a study tool. It completes the series' 8-day unique palette sequence.

Rate Before Reveal, Reveal Before Rate

Rating buttons are disabled until the card is flipped. This enforces the correct learning mechanic: the user must attempt to recall the answer before seeing it, which is the psychological basis of active recall testing.

Build Log — July 5, 2026

Time	Action	Decision
T+0:00	Idea selection	Flashcards — combines AI generation with an established learning science algorithm (SM-2)
T+0:15	Card flip design	Pure CSS 3D transform — no GSAP, no requestAnimationFrame; 4 CSS rules sufficient
T+0:35	SM-2 implementation	Simplified to ease/interval/ reps/ due; full SuperMemo variant out of scope for single file
T+0:55	Rating UX	Buttons disabled until flip — enforces active recall; enables rating only after answer revealed
T+1:15	Generation prompt	count + difficulty injected into prompt; JSON array output; fallback empty deck guard
T+1:35	Mastery metric	reps>=3 && ease>=2.3 as mastered threshold; surfaced in deck stats row

T+1:55	Due queue logic	Date.now() ms timestamps for due dates; fallback to first 10 for new decks
T+2:10	GitHub + LinkedIn	Links added to PDF cover per series standing instruction
T+2:20	Documentation	Whitepaper, Source Code Paper, History Paper compiled into single PDF

Key Decisions Rejected

Rejected	Why Rejected
Full SuperMemo SM-2	Matrix-based optimal intervals require lookup tables; simplified version produces equivalent sub
FSRS algorithm	More accurate than SM-2 but requires floating-point parameter fitting from review history; overk
Deck import/export (Anki .apkg)	SQLite-based .apkg format requires a WASM SQLite parser; deferred to Pro tier
Image cards	Vision model dependency adds setup friction; text-only cards cover 90% of study use cases
Swipe gestures for rating	Swipe conflict with scroll on mobile; tap buttons are more reliable and explicit

What Day 9 Could Build On

- Streak tracking and daily review reminders via Web Notifications API
- Anki .apkg import via WASM SQLite
- Image card support for visual subjects
- Per-deck performance graphs using Canvas 2D
- Multiple choice mode alongside open Q&A;
- Collaborative deck sharing via QR code or URL export

FlashMind — Day 08 of the Daily Local LLM Startup Series

Built by Mayank Swaraj | Mayank Creations | July 5, 2026

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